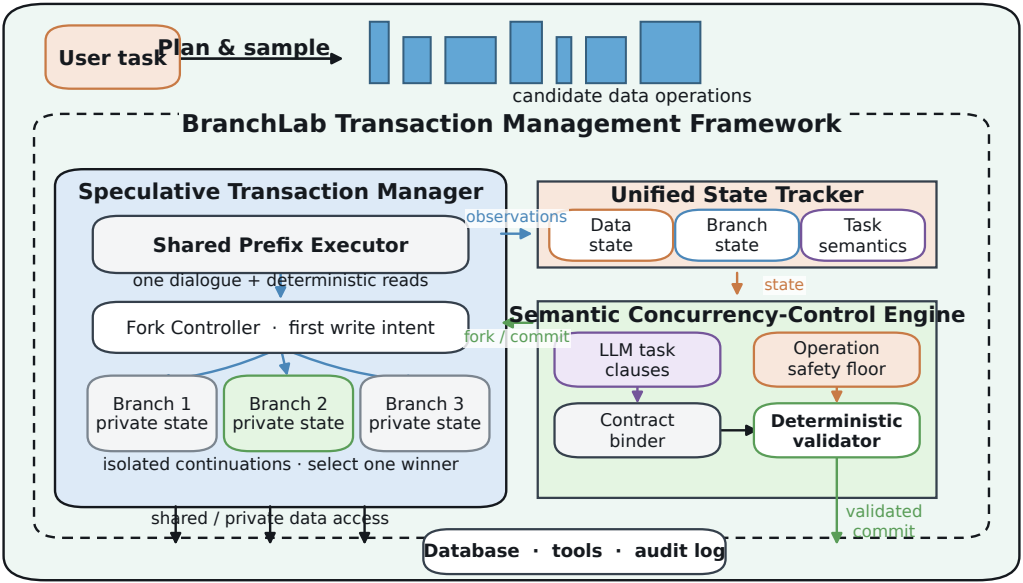

User task
Plan & sample
candidate data operations
BranchLab Transaction Management Framework
Speculative Transaction Manager
Shared Prefix Executor
one dialogue + deterministic reads
Fork Controller · first write intent
Branch 1 private state
Branch 2 private state
Branch 3 private state
isolated continuations · select one winner
shared / private data access
observations
fork / commit
Unified State Tracker
Data state
Branch state
Task semantics
state
Semantic Concurrency-Control Engine
LLM task clauses
Operation safety floor
Contract binder
Deterministic validator
validated commit
Database · tools · audit log